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Clock frequency regulating circuit

Abstract

A clock frequency regulating circuit includes a memory circuit, a frequency detection circuit, and a clock management unit (CMU). The memory circuit is configured to store data according to an input clock, and to output the data according to an output clock. The frequency detection circuit is coupled with the memory circuit, and configured to detect the variation in a used amount of a storage space of the memory circuit and accordingly generate a frequency detection result. The CMU is coupled with the frequency detection circuit, and configured to regulate the frequency of the output clock according to the frequency detection result.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present disclosure relates to a clock frequency regulating circuit, especially to a clock frequency regulating circuit capable of adaptively regulating the frequency of an output clock of a memory circuit according to the variation in the used amount of a storage space of the memory circuit.

2. Description of Related Art

[0002] A signal will attenuate after long distance transmission, and a retimer can receive the attenuated signal and recover it. The retimer is a mixed-signal analog/digital device having a protocol-aware capability and being able to extract a clock of a received signal; accordingly, the retimer can recover data of the received signal according to the extracted clock and use a clean version of the extracted clock to transmit a fresh copy of the data. In brief, the retimer can completely recover data of a received signal and transmit a fresh copy of the data.

[0003] Generally, a conventional retimer must align a transmission clock (i.e., an output clock) with a reception clock (i.e., an input clock) to realize frequency synchronization and thereby reduce the size of a buffer, that is used for compensating for the reception/transmission clock frequency deviation, and consequently miniaturize the area of the whole circuit.

[0004] However, in the process of aligning the transmission clock, if the difference between the frequency of the transmission clock and the frequency of the reception clock is significant, a large amount of frequency variation in the transmission clock will occur at the moment of adjusting the transmission clock, and this will prevent a back-end reception device from correctly receiving the data outputted by the retimer. In addition, aligning the transmission clock with the reception clock will cause the output signal of the retimer to carry a frequency fluctuation of the input signal of the retimer, which will degrade the quality of the output signal.

SUMMARY OF THE INVENTION

[0005] An objective of the present disclosure is to disclose a clock frequency regulating circuit applicable to a retimer and capable of preventing the problems of the prior art.

[0006] An embodiment of the present disclosure of the clock frequency regulating circuit includes a memory circuit, a frequency detection circuit, and a clock management unit (CMU). The memory circuit is configured to store data according to an input clock and output the data according to an output clock. The frequency detection circuit is coupled with the memory circuit, and is configured to detect a variation in a used amount of a storage space of the memory circuit and accordingly generate a frequency detection result. The CMU is coupled with the frequency detection circuit, and is configured to regulate a frequency of the output clock according to the frequency detection result. In an exemplary implementation of the embodiment, the frequency detection circuit determines which of a plurality of variation intervals the variation in the used amount of the storage space of the memory circuit falls within; when the variation falls within a first variation interval of the plurality of variation intervals, the frequency detection circuit generates the frequency detection result to request the CMU to regulate the frequency of the output clock according to a first degree of adjustment; and when the variation falls within a second variation interval of the plurality of variation intervals, the frequency detection circuit generates the frequency detection result to request the CMU to regulate the frequency of the output clock according to a second degree of adjustment.

[0007] These and other objectives of the present invention will no doubt become obvious to those

of ordinary skill in the art after reading the following detailed description of the preferred embodiments that are illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 shows an embodiment of the clock frequency regulating circuit of the present disclosure.

[0009] FIG. 2 shows the variation in the deviation of the frequency of the input clock of FIG. 1, the adjustment in the frequency of the output clock of FIG. 1, and the variation in the water level of the memory circuit of FIG. 1.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0010] The present specification discloses a clock frequency regulating circuit applicable to a retimer and capable of adaptively regulating the frequency of an output clock of a memory circuit according to the variation in the used amount of a storage space of the memory circuit.

[0011] FIG. 1 shows an embodiment of the clock frequency regulating circuit of the present disclosure. As shown in FIG. 1, the clock frequency regulating circuit **100** includes a memory circuit **110**, a frequency detection circuit **120**, a water-level detection circuit **130**, and a clock management unit (CMU) **140**.

[0012] Referring to FIG. 1, the memory circuit **110** is a circuit having a storage function such as a static random-access memory (SRAM), a first-in-first-out buffer (FIFO), or a counter. The memory circuit **110** is configured to store data according to an input clock CLK.sub.IN and output the data according to an output clock CLK.sub.OUT. The frequency detection circuit **120** is coupled with the memory circuit **110** and the CMU **140**, and is configured to detect the variation in the storage space (not shown in FIG. 1) of the memory circuit **110** (e.g., the variation in the used amount of the storage space in a predetermined period of time) and thereby generate a frequency detection result DET.sub.FREQ for the CMU **140**. The water-level detection circuit **130** is coupled with the memory circuit **110** and the CMU **140**, and is configured to detect the used amount of the storage space (hereinafter referred to as “water level”) of the memory circuit **110** and thereby generate a water-level detection result DET.sub.WL. The CMU **140** is coupled with the frequency detection circuit **120** and the water-level detection circuit **130**, and is configured to regulate the frequency of the output clock CLK.sub.OUT according to the frequency detection result DET.sub.FREQ and optionally according to the water-level detection result DET.sub.WL. It is noted that the term “water level” in this specification can be understood as the remaining amount of the storage space. It is also noted that the water-level detection circuit **130** can be disabled or omitted from the present embodiment if the performance of the clock frequency regulating circuit **100** is negligible.

[0013] FIG. 2 shows the variation in the deviation of the frequency of the input clock CLK.sub.IN, the adjustment in the frequency of the output clock CLK.sub.OUT, and the variation in the water level of the memory circuit **110**, wherein the unit of the variation in the deviation of the frequency of the input clock CLK.sub.IN and the unit of the adjustment in the frequency of the output clock CLK.sub.OUT are parts per million (ppm), and such unit (i.e., ppm) can be used to show the deviation from a specific central frequency. As shown in FIG. 2, the deviation in the frequency of the input clock CLK.sub.IN varies periodically: before a time point T.sub.0, the water level of the memory circuit **110** is zero (or alternatively, a non-zero specific water level); after the time point T.sub.0, the water level varies with the input clock CLK.sub.IN and the output clock CLK.sub.OUT; after the time point T.sub.0, the deviation in the frequency of the output clock CLK.sub.OUT varies with the water-level to make the water level converge toward a predetermined water level. Generally, when the frequency of the output clock CLK.sub.OUT gets higher, which implies that the data output rate of the memory circuit **110** gets higher, the water

level gradually drops; and when the frequency of the output clock CLK.sub.OUT gets lower, the water level gradually rises.

[0014] Referring to FIG. 2, the time points T.sub.0~T.sub.10 jointly defines ten time intervals as follows: T.sub.0~T.sub.1, T.sub.1~T.sub.2, . . . , T.sub.8~T.sub.9, and T.sub.9~T.sub.10. In each time interval, the water level of the memory circuit **110** varies with time; to be more specific, the water level at the start of a time interval (e.g., T.sub.0 of the time interval T.sub.0~T.sub.1) and the water level at the end of the same time interval (e.g., T.sub.1 of the time interval T.sub.0~T.sub.1) can jointly define the variation in the water level in this time interval. The water level at the end of a time interval may represent the water level of this time interval, but the implementation of the present invention is not limited thereto.

[0015] Referring to Tables 1~3 below, these tables show how the clock frequency regulating circuit **100** regulates the frequency of the output clock CLK.sub.OUT according to the variation in the water level of the memory circuit **110** and further according to the water level of the memory circuit **110**.

TABLE-US-00001 TABLE 1 Variation in water level F1 F2 F3 F4 F5 F6 F7 F8 WL.sub.NOW – >2000 ppm >1000 ppm >500 ppm >250 ppm <250 ppm <500 ppm <1000 ppm <2000 ppm WL.sub.BEFORE (variation interval) DET.sub.FREQ +1000 ppm +500 ppm +250 ppm +125 ppm –125 ppm –250 ppm –500 ppm –1000 ppm

TABLE-US-00002 TABLE 2 Water level WL1 WL2 WL3 WL4 Ragne of 100%~75% 75%~50% 50%~25% 25%~0% water level (water-level interval) DET.sub.WL +250 ppm +125 ppm –125 ppm –250 ppm

TABLE-US-00003 TABLE 3 Time point (microsecond) T1 T2 T3 T4 T5 T6 T7 T8 T9 T10 Water level WL3 WL2 WL2 WL2 WL3 WL4 WL4 WL3 WL2 WL2 Variation in zero F3 F4 F5 F6 F5 F4 F4 F3 F4 water level Total –125 ppm +375 ppm +250 ppm 0 ppm –375 ppm –375 pp –125 ppm 0 ppm +375 ppm +250 ppm adjustment in frequency of CLK.sub.OUT

[0016] Referring to Table 1 and FIG. 2, when the frequency detection circuit **120** determines that “the variation in the water level of the memory circuit **110** in a time interval” (e.g., the water level at the end of a time interval (hereinafter referred to as “WL.sub.NOW”) minus the water level at the start of the same time interval (hereinafter referred to as “WL.sub.BEFORE”); or multiplying a weighting by the result of subtracting WL.sub.BEFORE from WL.sub.NOW, wherein the weighting can be determined according to the last/previous frequency detection result DET.sub.FREQ) is higher than a threshold (i.e., >2000 ppm, >1000 ppm, >500 ppm, or >250 ppm as shown in Table 1), the frequency detection circuit **120** generates a corresponding frequency detection result DET.sub.FREQ (i.e., F1, F2, F3, or F4 as shown in Table 1) to request the CMU **140** to increase the frequency of the output clock CLK.sub.OUT by a predetermined degree of adjustment (i.e., +1000 ppm, +500 ppm, +250 ppm, or +125 ppm as shown in Table 1). On the other hand, when the frequency detection circuit **120** determines that “the variation in the water level of the memory circuit **110** in a time interval” (e.g., the result of subtracting WL.sub.BEFORE from WL.sub.NOW; or multiplying a weighting by the result of subtracting WL.sub.BEFORE from WL.sub.NOW, wherein the weighting can be determined according to the last/previous frequency detection result DET.sub.FREQ) is lower than a threshold (i.e., <250 ppm, <500 ppm, <1000 ppm, or <2000 ppm as shown in Table 1), the frequency detection circuit **120** generates a corresponding frequency detection result DET.sub.FREQ (i.e., F5, F6, F7, or F8 as shown in Table 1) to request the CMU **140** to decrease the frequency of the output clock CLK.sub.OUT by a predetermined degree of adjustment (i.e., –125 ppm, –250 ppm, –500 ppm, or –1000 ppm as shown in Table 1).

[0017] Referring to Table 1, it is noted that the aforementioned thresholds jointly define a plurality of variation intervals as follows: >2000 ppm, >1000 ppm, . . . , <1000 ppm, and <2000 ppm. The frequency detection circuit **120** can determine which of the plurality of variation intervals the variation in the water level falls within in the order of these variation intervals (e.g., an order from the first variation interval “>2000 ppm” to the last variation interval “<2000 ppm”). It is also noted

that more or less variation intervals can be applied according to implementation needs; additionally, the degree of adjustment applied to each variation interval can be determined according to implementation needs.

[0018] Referring to Table 2 and FIG. 2, when the water-level detection circuit **130** determines that the water level

[00001]($e.g., WL_{NOW}, \text{ or } \frac{WL_{BEFORE} + WL_{NOW}}{2}$)

of the memory circuit **110** in a time interval falls within a water-level interval (e.g., 100%~75% or 75%~50% as shown in Table 2) of a higher water level, the water-level detection circuit **130** generates a corresponding water-level detection result DET.sub.WL (e.g., WL1 or WL2 as shown in Table 2) to request the CMU **140** to increase the frequency of the output clock CLK.sub.OUT by a degree of adjustment (e.g., +250 ppm or +125 ppm as shown in Table 2). Similarly, when the water-level detection circuit **130** determines that the water level

[00002]($e.g., WL_{NOW}, \text{ or } \frac{WL_{BEFORE} + WL_{NOW}}{2}$)

the memory circuit **110** in a time interval falls within a water-level interval (e.g., 50%~25% or 25%~0% as shown in Table 2) of a lower water level, the water-level detection circuit **130** generates a corresponding water-level detection result DET.sub.WL (e.g., WL3 or WL4 as shown in Table 2) to request the CMU **140** to decrease the frequency of the output clock CLK.sub.OUT by a degree of adjustment (e.g., -125 ppm or -250 ppm as shown in Table 2).

[0019] It is noted that more or less water-level intervals can be applied according to implementation needs; additionally, the degree of adjustment applied to each water-level interval can be determined according to implementation needs. It is also noted that the total number of the water-level intervals (e.g., the four water-level intervals as shown in Table 2) is less than the total number of the aforementioned variation intervals (e.g., the eight variation intervals as shown in Table 1), but the implementation of the present invention is not limited thereto.

[0020] Referring to Table 3, the CMU **140** regulates the frequency of the output clock CLK.sub.OUT according to the frequency detection result DET.sub.FREQ of Table 1 and the water-level detection result DET.sub.WL of Table 2. In detail: [0021] (1) at the time point T.sub.1: the frequency detection circuit **120** takes the last/previous water level (WL.sub.BEFORE) as the water level (WL.sub.NOW) at T.sub.1 and therefore does not request the CMU **140** to regulate the frequency of the output clock CLK.sub.OUT because “WL.sub.NOW-WL.sub.BEFORE=0”, or the frequency detection circuit **120** can request the CMU to regulate the frequency of the output clock CLK.sub.OUT according to a predetermined rule (e.g., according to the degree of a difference between the water level at the time point T.sub.1 and a predetermined water level), wherein the predetermined rule can be set according to design/implementation needs; in addition, the water-level detection circuit **130** determines that the water level at the time point T.sub.1 falls within a water-level interval 50%~25% and thereby generates a corresponding water-level detection result DET.sub.WL (i.e., WL3 in Table 2) to request the CMU **140** to regulate the frequency of the output clock CLK.sub.OUT by -125 ppm. [0022] (2) at the time point T.sub.2: the frequency detection circuit **120** determines that the variation in the water level is less than 1000 ppm but greater than 500 ppm and thereby generates a corresponding frequency detection result DET.sub.FREQ (i.e., F3 in Table 1) to request the CMU **140** to regulate the frequency of the output clock CLK.sub.OUT by +250 ppm; in the meantime, the water-level detection circuit **130** determines that the water level at the time point T.sub.2 falls within a water-level interval 75%~50% and thereby generates a corresponding water-level detection result DET.sub.WL (i.e., WL2 in Table 2) to request the CMU **140** to regulate the frequency of the output clock CLK.sub.OUT by +125 ppm; as a result, the CMU **140** regulates the frequency of the output clock CLK.sub.OUT by +375 ppm (i.e., 250 ppm+125 ppm) according to the frequency detection result DET.sub.FREQ and the water-level detection result DET.sub.WL. [0023] (3) the ways to regulate the frequency of the output clock CLK.sub.OUT at the other time points T.sub.3~T.sub.10 can be derived from the above description based on Tables 1~3 and FIG. 2.

[0024] Referring to FIGS. 1~2, in an embodiment: after the memory circuit **110** stores the data for a first period of time according to the input clock CLK.sub.IN, the frequency detection circuit **120** starts detecting the variation in the storage space of the memory circuit **110**, wherein the first period of time can be determined according to design/implementation needs. In an embodiment: after the memory circuit **110** stores the data for a second period of time according to the input clock CLK.sub.IN, the water-level detection circuit **130** starts detecting the water level of the memory circuit **110**, wherein the second period of time can be determined according to design/implementation needs. In an embodiment, the frequency detection circuit **120** and/or the water-level detection circuit **130** optionally start(s) performing the detection operation when the memory circuit **110** starts storing the data (i.e., at the time point T START as shown in FIG. 2) according to the input clock CLK.sub.IN. It is noted that if the time intervals for sampling the water level (i.e., T.sub.0~T.sub.1, T.sub.1~T.sub.2, T.sub.2~T.sub.3, and so on) are far shorter than the cycle of the input clock CLK.sub.IN, the cycle of the output clock CLK.sub.OUT will be equal or similar to the cycle of the input clock CLK.sub.IN, but the implementation of the present invention is not limited thereto.

[0025] It is noted that people having ordinary skill in the art can selectively use some or all of the features of any embodiment in this specification or selectively use some or all of the features of multiple embodiments in this specification to implement the present invention as long as such implementation is practicable; in other words, the way to implement the present invention can be flexible based on the present disclosure.

[0026] To sum up, the clock frequency regulating circuit of the present disclosure can regulate the frequency of an output clock of a memory circuit according to the variation in the used amount of a storage space of the memory circuit and can be applied to a retimer or any other device in want of the function of clock frequency regulation.

[0027] The aforementioned descriptions represent merely the preferred embodiments of the present invention, without any intention to limit the scope of the present invention thereto. Various equivalent changes, alterations, or modifications based on the claims of the present invention are all consequently viewed as being embraced by the scope of the present invention.

Claims

1. A clock frequency regulating circuit, comprising: a memory circuit configured to store data according to an input clock and output the data according to an output clock; a frequency detection circuit coupled with the memory circuit, the frequency detection circuit configured to detect a variation in a used amount of a storage space of the memory circuit and accordingly generate a frequency detection result; and a clock management unit (CMU) coupled with the frequency detection circuit, the CMU configured to regulate a frequency of the output clock according to the frequency detection result.
2. The clock frequency regulating circuit of claim 1, wherein after the memory circuit stores the data for a period of time according to the input clock, the frequency detection circuit starts detecting the variation in the used amount of the storage space of the memory circuit.
3. The clock frequency regulating circuit of claim 1, wherein the frequency detection circuit determines which of a plurality of variation intervals the variation falls within; when the variation falls within a first variation interval of the plurality of variation intervals, the frequency detection circuit generates the frequency detection result to request the CMU to regulate the frequency of the output clock according to a first degree of adjustment; and when the variation falls within a second variation interval of the plurality of variation intervals, the frequency detection circuit generates the frequency detection result to request the CMU to regulate the frequency of the output clock according to a second degree of adjustment.
4. The clock frequency regulating circuit of claim 3, wherein when the variation falls within the

first variation interval, the frequency detection circuit generates the frequency detection result to request the CMU to increase the frequency of the output clock according to the first degree of adjustment; and when the variation falls within the second variation interval, the frequency detection circuit generates the frequency detection result to request the CMU to decrease the frequency of the output clock according to the second degree of adjustment.

5. The clock frequency regulating circuit of claim 3, wherein the first degree of adjustment is different from the second degree of adjustment.

6. The clock frequency regulating circuit of claim 3, further comprising: a water-level detection circuit coupled to the memory circuit and the CMU, the water-level detection circuit configured to detect the used amount of the storage space of the memory circuit and thereby generate a water-level detection result, wherein the CMU is further configured to regulate the frequency of the output clock according to the water-level detection result.

7. The clock frequency regulating circuit of claim 6, wherein after the memory circuit stores the data for a period of time according to the input clock, the water-level detection circuit starts detecting the used amount of the storage space of the memory circuit.

8. The clock frequency regulating circuit of claim 6, wherein the water-level detection circuit determines which of a plurality of water-level intervals the used amount falls within; when the used amount falls within a first water-level interval of the plurality of water-level intervals, the water-level detection circuit generates the water-level detection result to request the CMU to regulate the frequency of the output clock according to a third degree of adjustment; and when the used amount falls within a second water-level interval of the plurality of water-level intervals, the water-level detection circuit generates the water-level detection result to request the CMU to regulate the frequency of the output clock according to a fourth degree of adjustment.

9. The clock frequency regulating circuit of claim 8, wherein when the used amount falls within the first water-level interval, the water-level detection circuit generates the water-level detection result to request the CMU to increase the frequency of the output clock according to the third degree of adjustment; and when the used amount falls within the second water-level interval, the water-level detection circuit generates the water-level detection result to request the CMU to decrease the frequency of the output clock according to the fourth degree of adjustment.

10. The clock frequency regulating circuit of claim 8, wherein the third degree of adjustment is different from the fourth degree of adjustment.

11. The clock frequency regulating circuit of claim 8, wherein a total number of the plurality of water-level intervals is less than a total number of the plurality of variation intervals.

12. The clock frequency regulating circuit of claim 1, further comprising: a water-level detection circuit coupled to the memory circuit and the CMU, the water-level detection circuit configured to detect the used amount of the storage space of the memory circuit and thereby generate a water-level detection result, wherein the CMU is further configured to regulate the frequency of the output clock according to the water-level detection result.

13. The clock frequency regulating circuit of claim 12, wherein after the memory circuit stores the data for a period of time according to the input clock, the water-level detection circuit starts detecting the used amount of the storage space of the memory circuit.

14. The clock frequency regulating circuit of claim 12, wherein the water-level detection circuit determines which of a plurality of water-level intervals the used amount falls within; when the used amount falls within a first water-level interval of the plurality of water-level intervals, the water-level detection circuit generates the water-level detection result to request the CMU to regulate the frequency of the output clock according to a first degree of adjustment; and when the used amount falls within a second water-level interval of the plurality of water-level intervals, the water-level detection circuit generates the water-level detection result to request the CMU to regulate the frequency of the output clock according to a second degree of adjustment.

15. The clock frequency regulating circuit of claim 14, wherein when the used amount falls within

the first water-level interval, the water-level detection circuit generates the water-level detection result to request the CMU to increase the frequency of the output clock according to the first degree of adjustment; and when the used amount falls within the second water-level interval, the water-level detection circuit generates the water-level detection result to request the CMU to decrease the frequency of the output clock according to the second degree of adjustment.

16. The clock frequency regulating circuit of claim 14, wherein the first degree of adjustment is different from the second degree of adjustment.

17. The clock frequency regulating circuit of claim 1, wherein the clock frequency regulating circuit is applied to a retimer.
